Chandrayaan Missions to the Moon

Introduction to the Chandrayaan Programme

Chandrayaan, a Sanskrit word meaning "Moon vehicle", is the name of India's series of lunar exploration missions run by the Indian Space Research Organisation (ISRO). The programme aims to study the Moon's surface, minerals, water, atmosphere and structure, and to demonstrate the technology of soft landing and roving on another world. So far, three missions have flown: Chandrayaan-1 in 2008, Chandrayaan-2 in 2019 and Chandrayaan-3 in 2023, and more are planned.

Chandrayaan-1 (2008)

Chandrayaan-1 was launched on 22 October 2008 from the Satish Dhawan Space Centre in Sriharikota on a PSLV-C11 rocket. It was India's first mission to the Moon. It consisted of an orbiter that circled the Moon at about 100 km altitude and a Moon Impact Probe (MIP), which separated and hit the lunar surface near the Shackleton crater at the south pole on 14 November 2008, making India the fourth country to place its flag on the Moon's surface. The spacecraft carried eleven scientific instruments, including five Indian instruments and others from NASA, the European Space Agency and Bulgaria. M. Annadurai, a scientist from Tamil Nadu, was the project director.

What Did Chandrayaan-1 Discover?

The most important discovery of Chandrayaan-1 was the presence of water molecules on the Moon. NASA's Moon Mineralogy Mapper (M3) instrument on the spacecraft detected water and hydroxyl molecules in the lunar surface soil, and the finding was announced in September 2009. The Moon Impact Probe also detected traces of water in the thin lunar atmosphere during its descent. Chandrayaan-1 also mapped minerals and chemical elements, produced high-resolution maps of the Moon's surface and helped reveal evidence of water ice in permanently shadowed craters near the poles. Communication with the spacecraft was lost in August 2009, but it had achieved most of its objectives.

Chandrayaan-2 (2019)

Chandrayaan-2 was launched on 22 July 2019 on a GSLV Mk III rocket (LVM3). It had an orbiter, a lander named Vikram and a rover named Pragyan. The orbiter was placed in lunar orbit successfully and continues to operate, sending valuable data. However, on 7 September 2019, during the final descent, the Vikram lander lost control and crashed near the south pole, because of a software issue and a braking problem. The failure taught ISRO important lessons, which were used to redesign the next mission.

Chandrayaan-3 (2023)

Chandrayaan-3 was launched on 14 July 2023 from Sriharikota on an LVM3-M4 rocket. It consisted of a propulsion module, a lander and a rover. After orbiting the Earth and then the Moon, the Vikram lander touched down on the Moon at about 6:04 pm Indian Standard Time on 23 August 2023, making India the fourth country, after the Soviet Union, the United States and China, to achieve a soft landing on the Moon, and the first to land near the lunar south pole.

Where Did Chandrayaan-3 Land?

Chandrayaan-3 landed near the south polar region of the Moon, at about 69.37 degrees south latitude and 32.32 degrees east longitude, on a plain between the craters Manzinus C and Simpelius N. The landing site was named Shiv Shakti Point. The south pole is of great scientific interest because permanently shadowed craters may hold water ice that could support future human missions.

Lander and Rover Names

The lander of Chandrayaan-3 was named Vikram, in honour of Dr. Vikram Sarabhai, the founder of India's space programme. The rover was named Pragyan, a Sanskrit word meaning wisdom. The Pragyan rover weighed about 26 kg, had six wheels, and travelled over a hundred metres on the lunar surface. It carried an Alpha Particle X-ray Spectrometer and a Laser-Induced Breakdown Spectroscope. The Vikram lander weighed about 1,700 kg including the rover and carried payloads to measure surface temperature (ChaSTE), moonquakes (ILSA) and plasma. The mission was designed to work for one lunar day, about 14 Earth days. P. Veeramuthuvel, an engineer from Villupuram in Tamil Nadu, was the project director, and S. Somanath was ISRO's chairman. The Moon-landing cost about Rs. 615 crore, which was lower than the budget of many science-fiction films.

Chandrayaan-3 Findings

The rover confirmed the presence of sulphur in the lunar south polar soil, and also detected aluminium, calcium, iron, chromium, titanium, manganese, silicon and oxygen. The ChaSTE probe measured a sharp temperature drop from about 50 degrees Celsius at the surface to about minus 10 degrees Celsius a few centimetres below. The lander also recorded a natural seismic event. In September 2023, the Vikram lander performed a "hop experiment", lifting off and landing about 40 cm away, as a test for future sample-return missions. The government declared 23 August as National Space Day to commemorate the success.

The Future: Chandrayaan-4 and Beyond

Chandrayaan-4 is planned as a lunar sample-return mission, approved in September 2024, in which a spacecraft will land, collect soil and bring it back to Earth. Chandrayaan-5, also known as LUPEX, is a joint mission with Japan's space agency JAXA to study water at the lunar south pole. India's long-term goal is to land an Indian astronaut on the Moon by 2040.

Challenges

Moon landings are difficult because of the long distance, the need for precise navigation, lunar dust, extreme temperature swings and surviving the freezing lunar night, which lasts about 14 Earth days.

Quick Facts

- Chandrayaan-1 launched on 22 October 2008 and helped confirm water molecules on the Moon.
- Chandrayaan-3 landed on 23 August 2023 near the lunar south pole, at Shiv Shakti Point.
- The lander was named Vikram and the rover was named Pragyan.
- India became the fourth country to soft-land on the Moon.
- 23 August is celebrated as National Space Day in India.